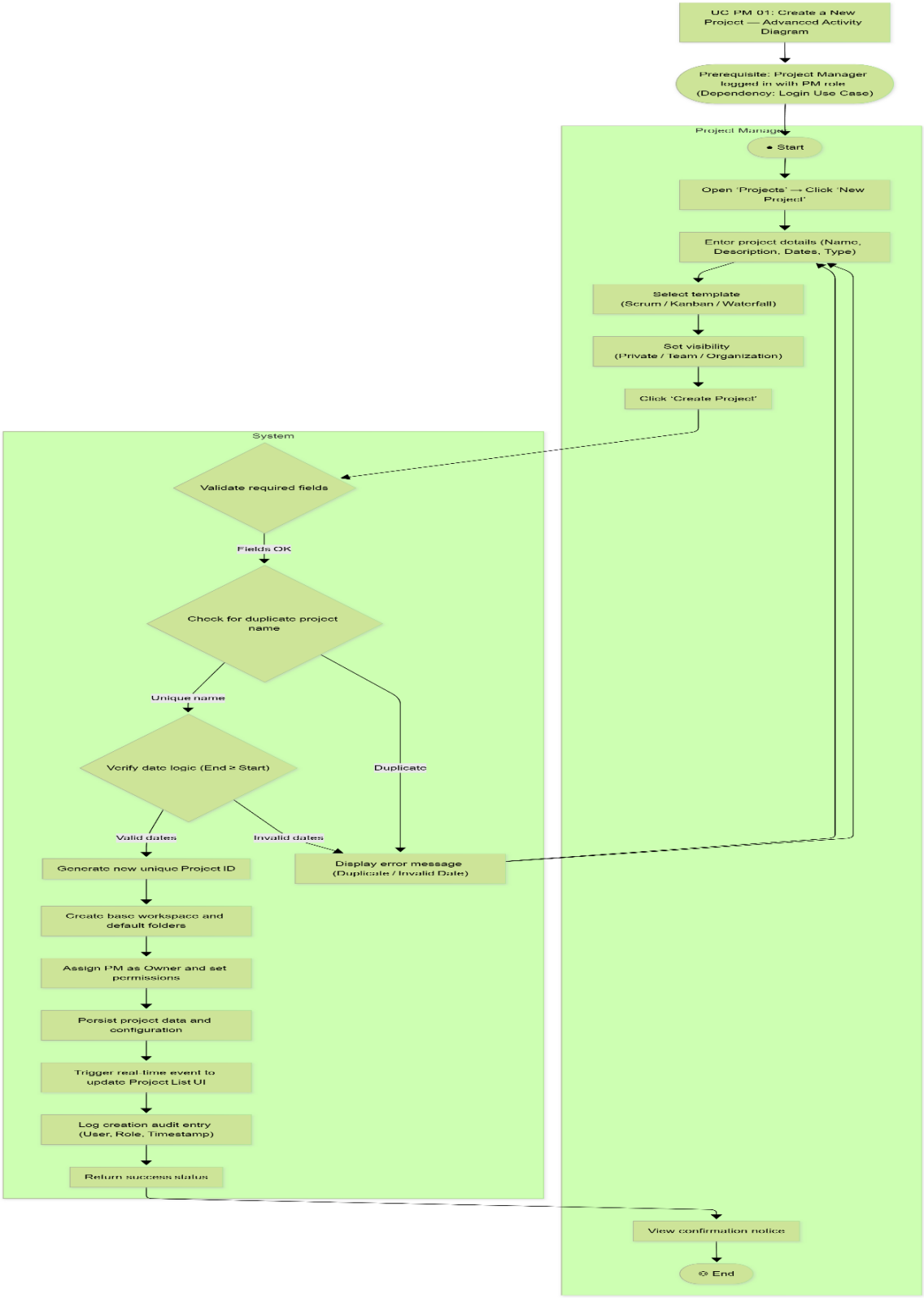
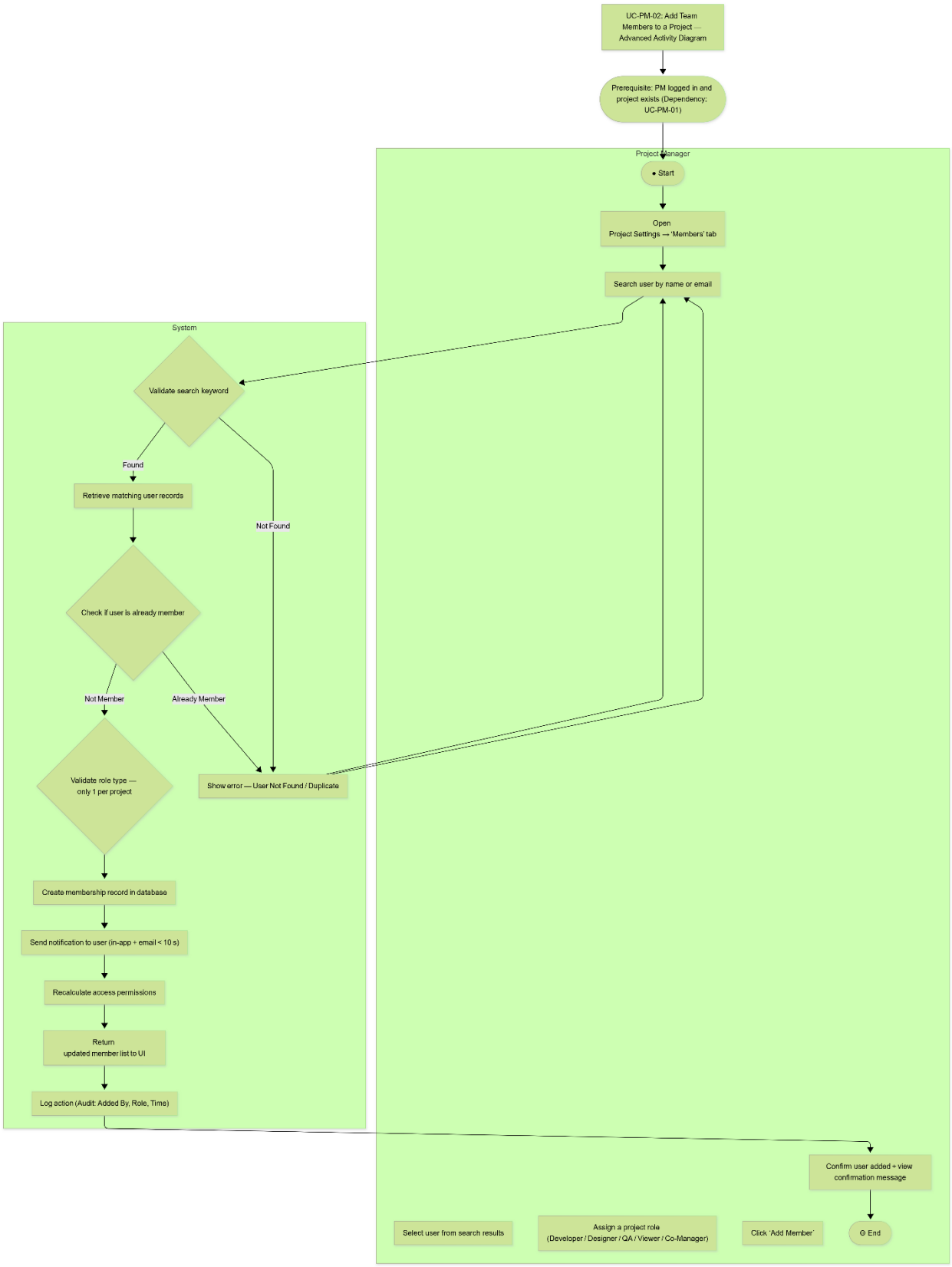
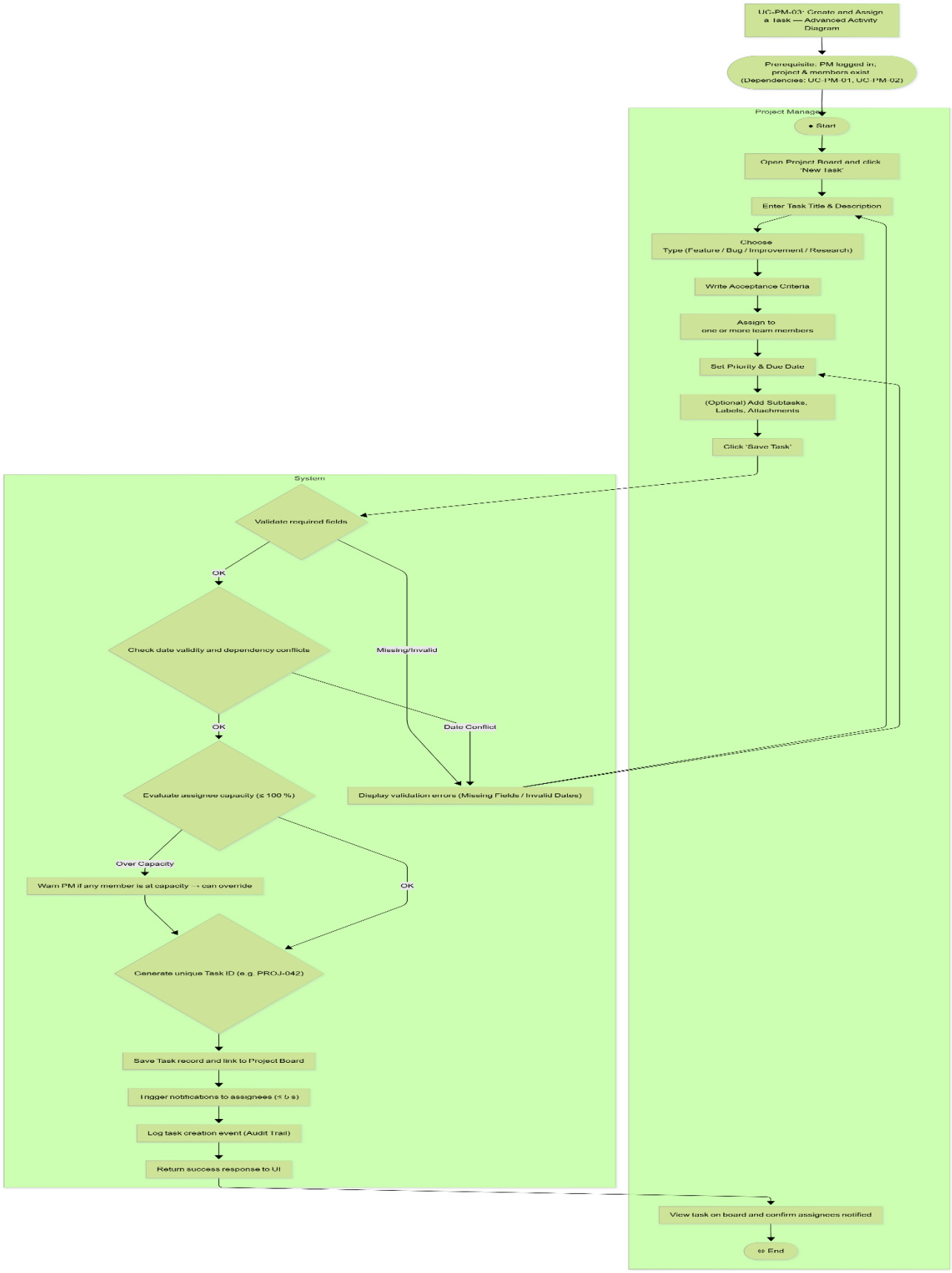
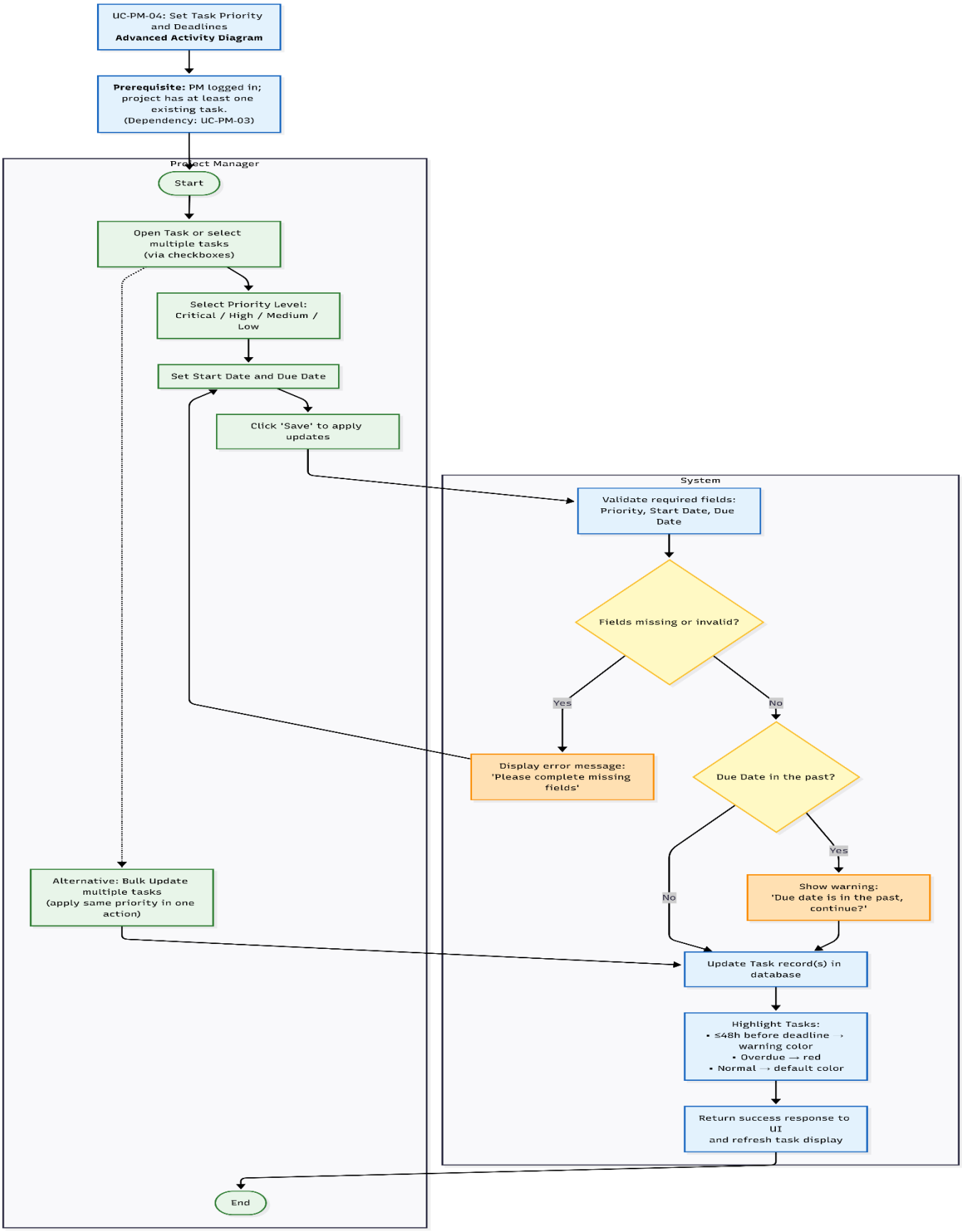
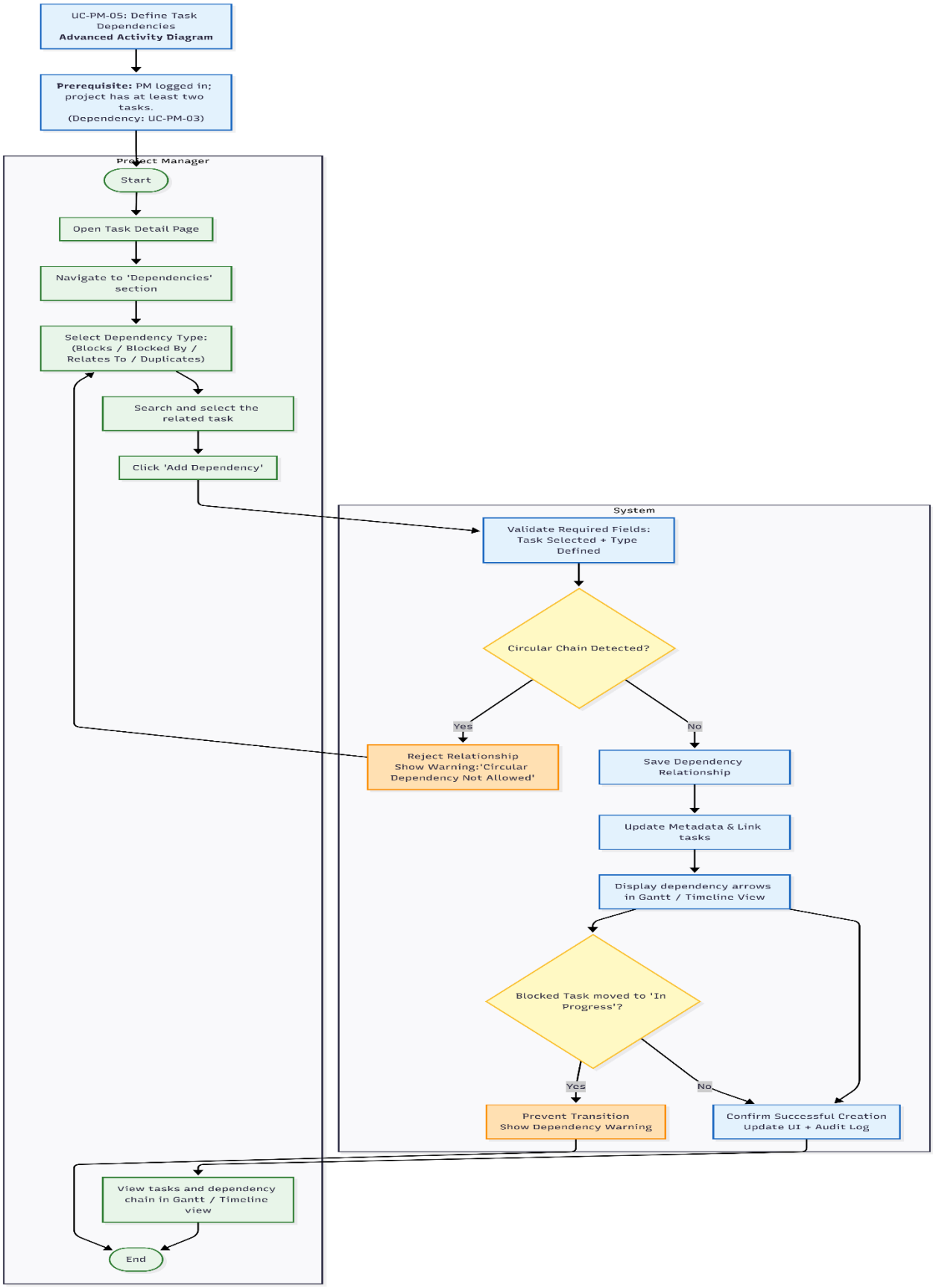
Activity diagrams for Project Managers

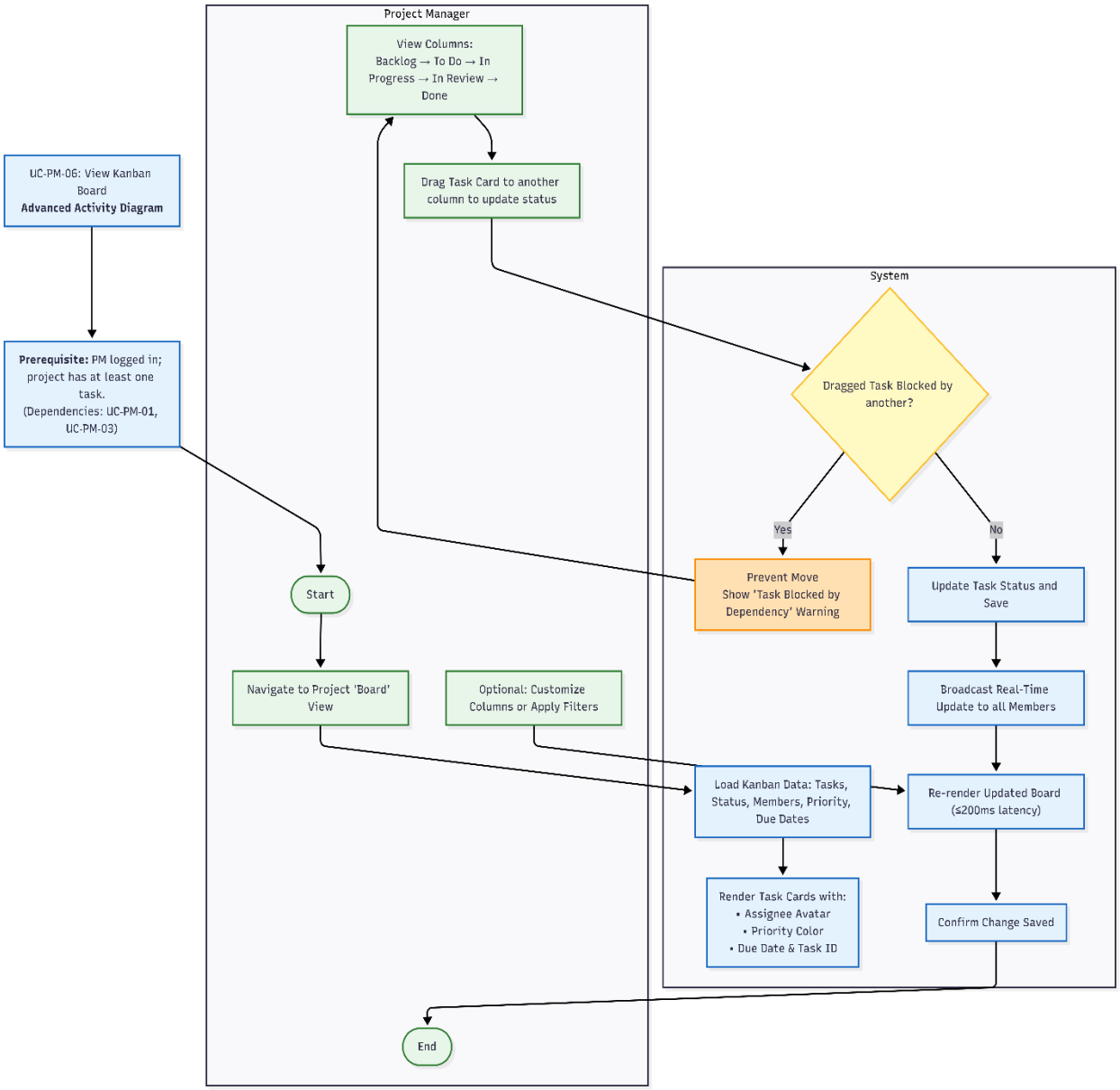


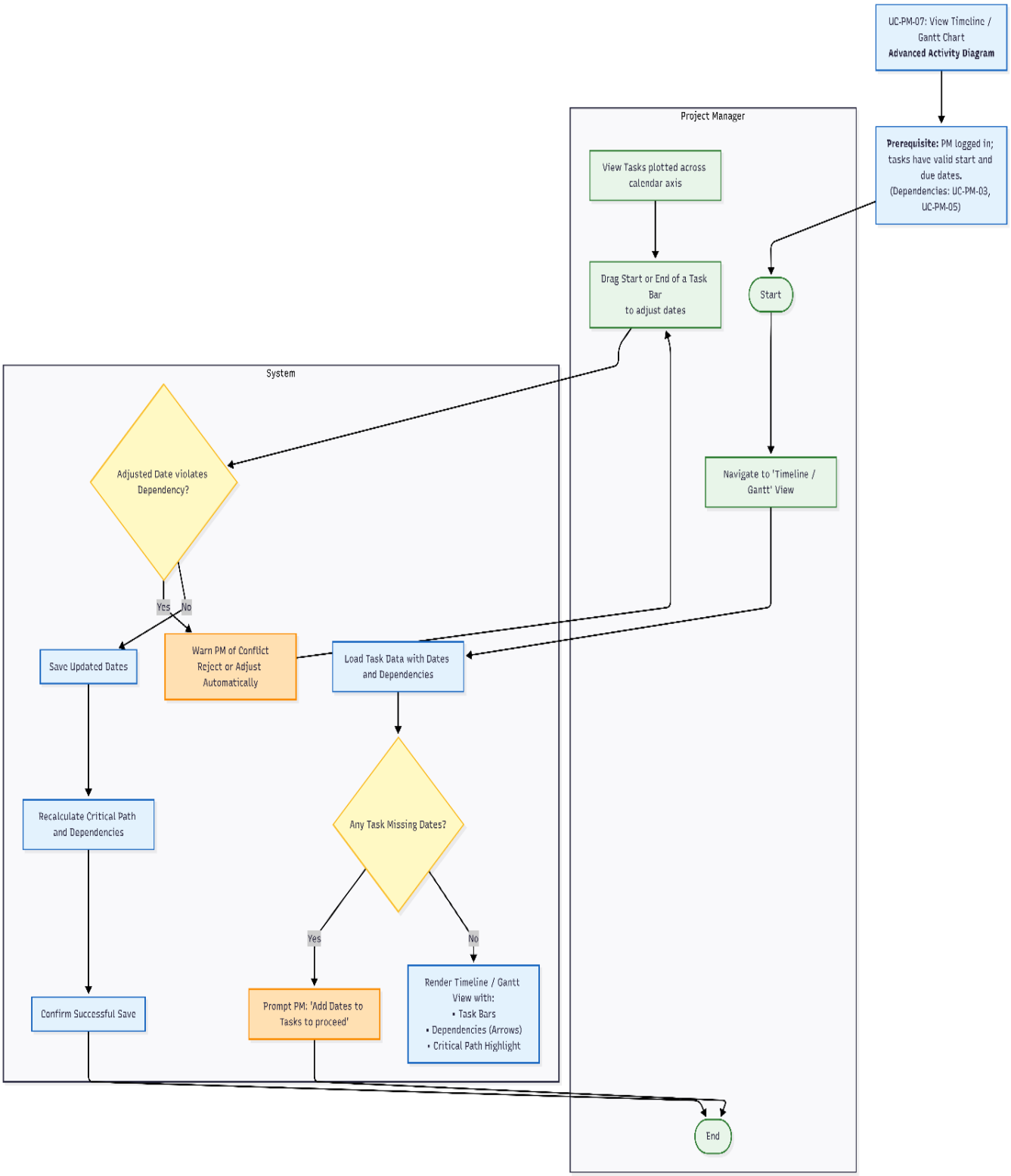

UC-PM-02: Add Team Members to a Project — Advanced Activity Diagram
Prerequisite: PM logged in and project exists (Dependency: UC-PM-01)
Project Manager
Start
Open Project Settings → 'Members' tab
Search user by name or email
System
Validate search keyword
Found
Not Found
Retrieve matching user records
Check if user is already member
Not Member
Already Member
Validate role type — only 1 per project
Show error — User Not Found / Duplicate
Create membership record in database
Send notification to user (in-app + email < 10 s)
Recalculate access permissions
Return updated member list to UI
Log action (Audit: Added By, Role, Time)
Confirm user added + view confirmation message
Select user from search results
Assign a project role (Developer / Designer / QA / Viewer / Co-Manager)
Click 'Add Member'
End

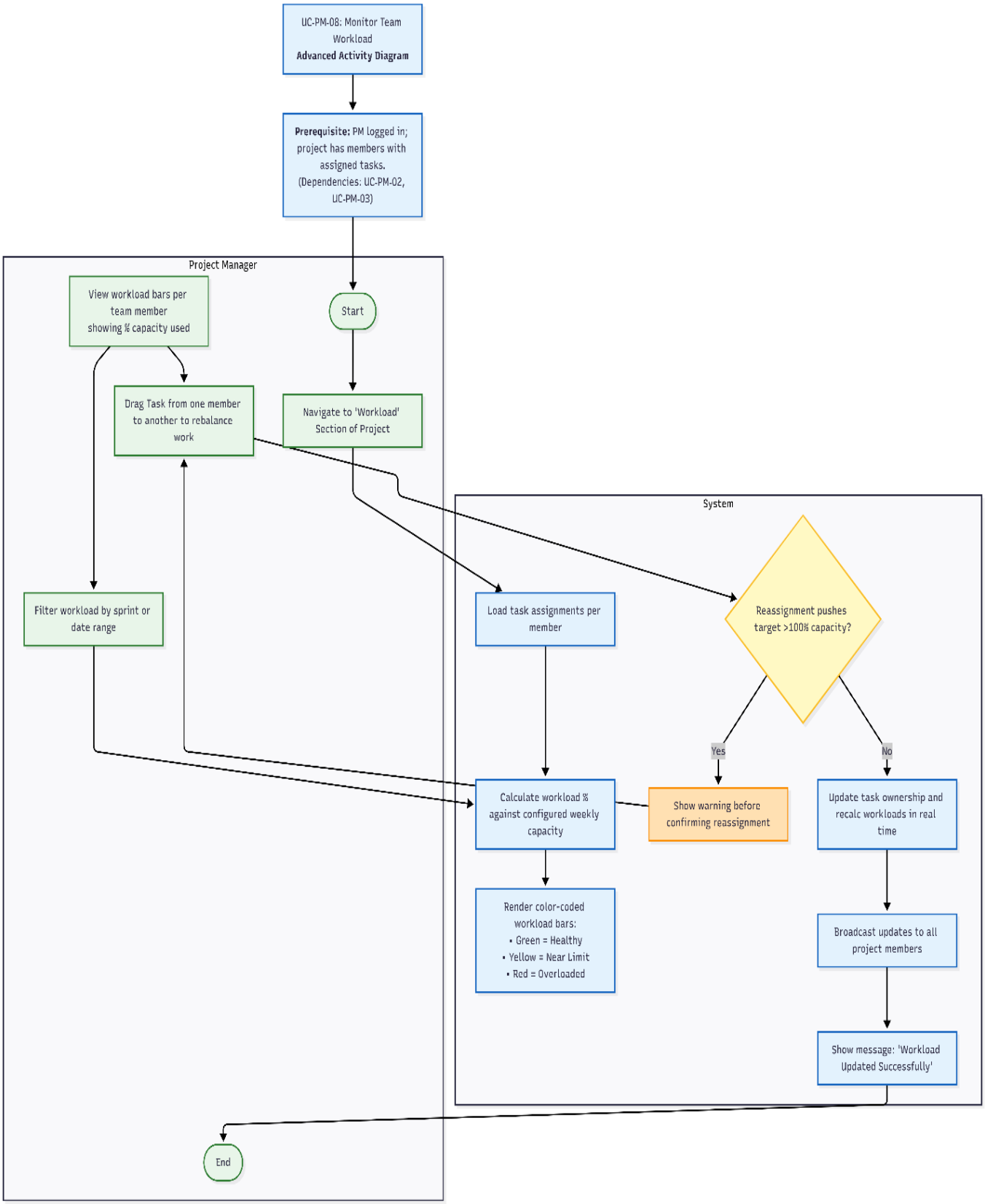

UC-PM-03: Create and Assign a Task — Advanced Activity Diagram
Prerequisite: PM logged in, project & members exist (Dependencies: UC-PM-01, UC-PM-02)
Project Manager
Start
Open Project Board and click 'New Task'
Enter Task Title & Description
Choose Type (Feature / Bug / Improvement / Research)
Write Acceptance Criteria
Assign to one or more team members
Set Priority & Due Date
(Optional) Add Subtasks, Labels, Attachments
Click 'Save Task'
System
Validate required fields
OK
Missing/Invalid
Check date validity and dependency conflicts
OK
Date Conflict
Evaluate assignee capacity (≤ 100 %)
Display validation errors (Missing Fields / Invalid Dates)
Over Capacity
Warn PM if any member is at capacity → can override
OK
Generate unique Task ID (e.g. PROJ-042)
Save Task record and link to Project Board
Trigger notifications to assignees (≤ 5 s)
Log task creation event (Audit Trail)
Return success response to UI
View task on board and confirm assignees notified
End

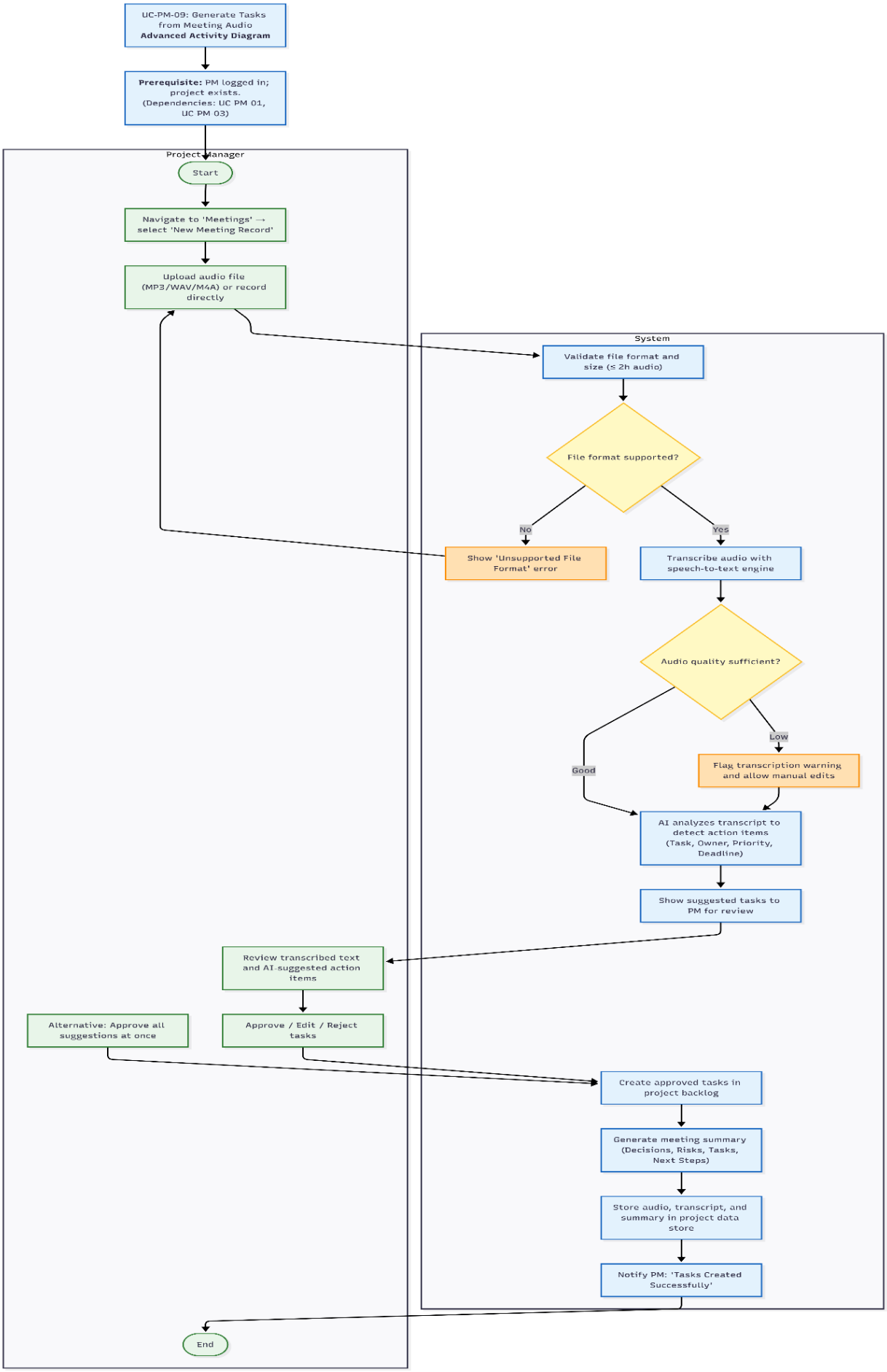

UC-PM-04: Set Task Priority and Deadlines
Advanced Activity Diagram
Prerequisite: PM logged in; project has at least one existing task. (Dependency: UC-PM-03)
Project Manager
Start
Open Task or select multiple tasks (via checkboxes)
Select Priority Level: Critical / High / Medium / Low
Set Start Date and Due Date
Click 'Save' to apply updates
Alternative: Bulk Update multiple tasks (apply same priority in one action)
End
System
Validate required fields: Priority, Start Date, Due Date
Fields missing or invalid?
Yes
No
Display error message: 'Please complete missing fields'
Due Date in the past?
Yes
No
Show warning: 'Due date is in the past, continue?'
Update Task record(s) in database
Highlight Tasks:
• ≤48h before deadline → warning color
• Overdue → red
• Normal → default color
Return success response to UI and refresh task display

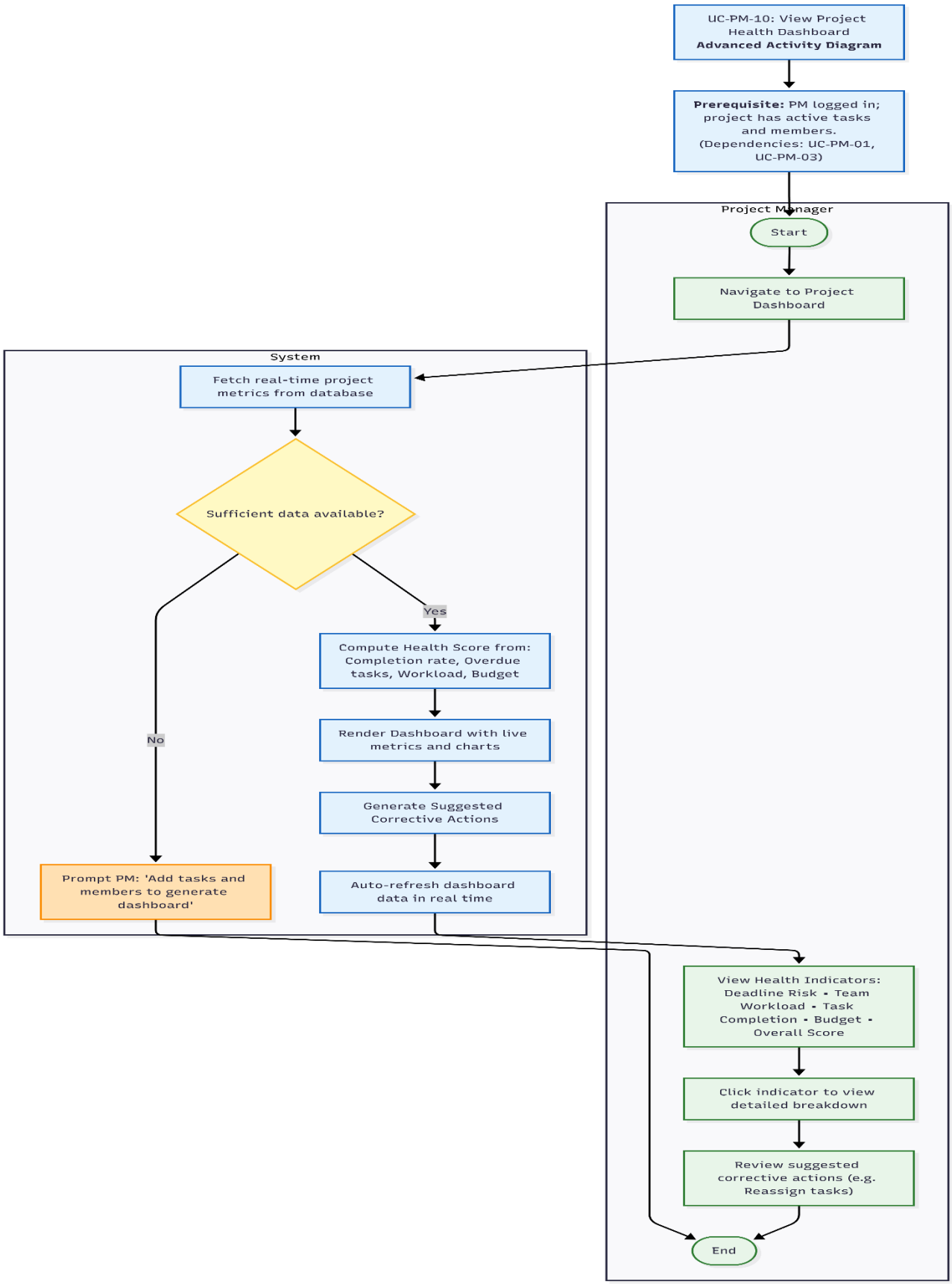

UC-PM-05: Define Task Dependencies
Advanced Activity Diagram
Prerequisite: PM logged in; project has at least two tasks.
(Dependency: UC-PM-03)
Project Manager
Start
Open Task Detail Page
Navigate to 'Dependencies' section
Select Dependency Type: (Blocks / Blocked By / Relates To / Duplicates)
Search and select the related task
Click 'Add Dependency'
System
Validate Required Fields: Task Selected + Type Defined
Circular Chain Detected?
Yes
No
Reject Relationship Show Warning:'Circular Dependency Not Allowed'
Save Dependency Relationship
Update Metadata & Link tasks
Display dependency arrows in Gantt / Timeline View
Blocked Task moved to 'In Progress'?
Yes
No
Prevent Transition Show Dependency Warning
Confirm Successful Creation Update UI + Audit Log
View tasks and dependency chain in Gantt / Timeline view
End

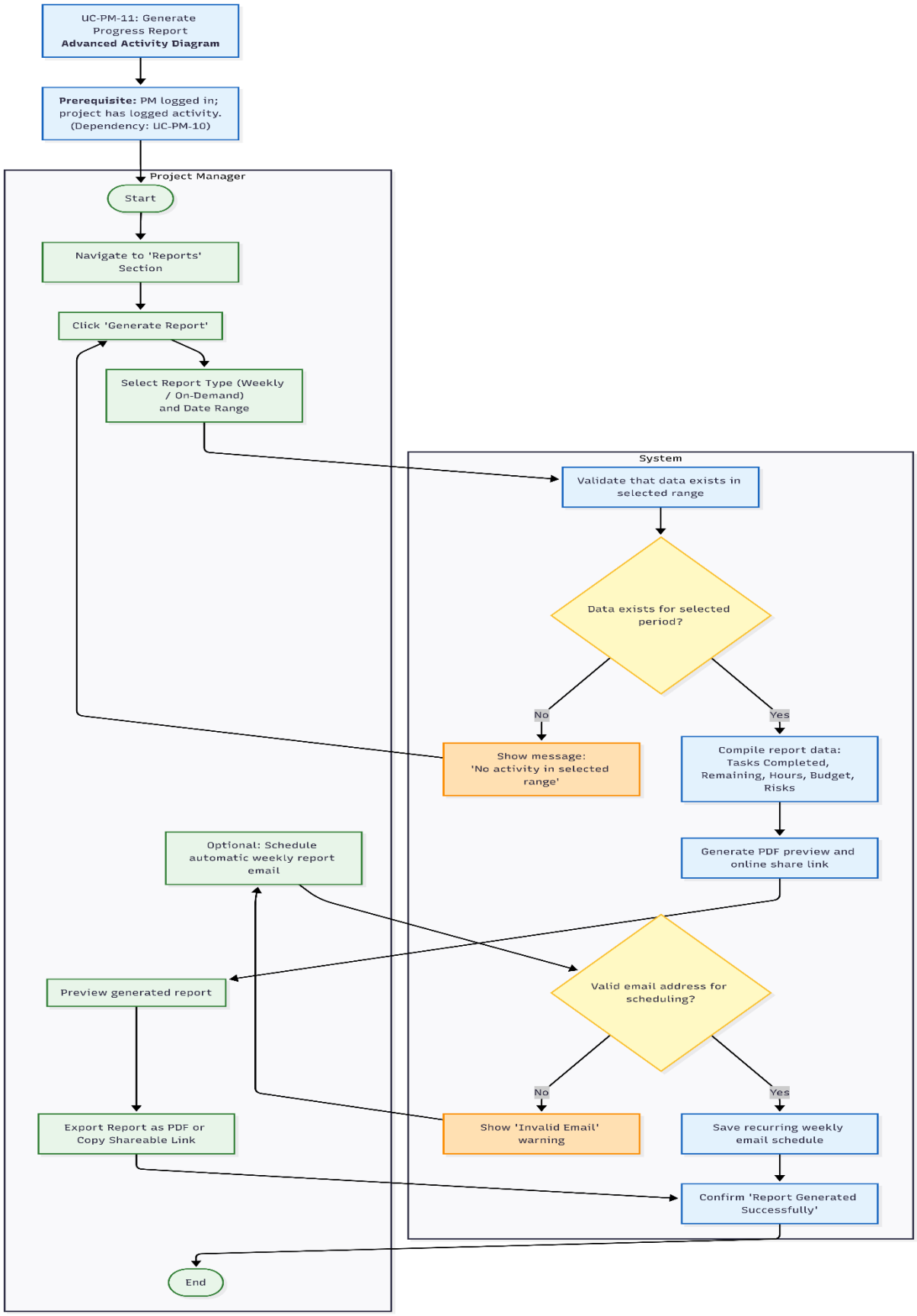

UC-PM-06: View Kanban Board
Advanced Activity Diagram
Prerequisite: PM logged in; project has at least one task. (Dependencies: UC-PM-01, UC-PM-03)
Project Manager
View Columns: Backlog → To Do → In Progress → In Review → Done
Drag Task Card to another column to update status
System
Dragged Task Blocked by another?
Yes
No
Prevent Move Show 'Task Blocked by Dependency' Warning
Update Task Status and Save
Broadcast Real-Time Update to all Members
Start
Navigate to Project 'Board' View
Optional: Customize Columns or Apply Filters
Load Kanban Data: Tasks, Status, Members, Priority, Due Dates
Re-render Updated Board (≤200ms latency)
Render Task Cards with:
• Assignee Avatar
• Priority Color
• Due Date & Task ID
Confirm Change Saved
End

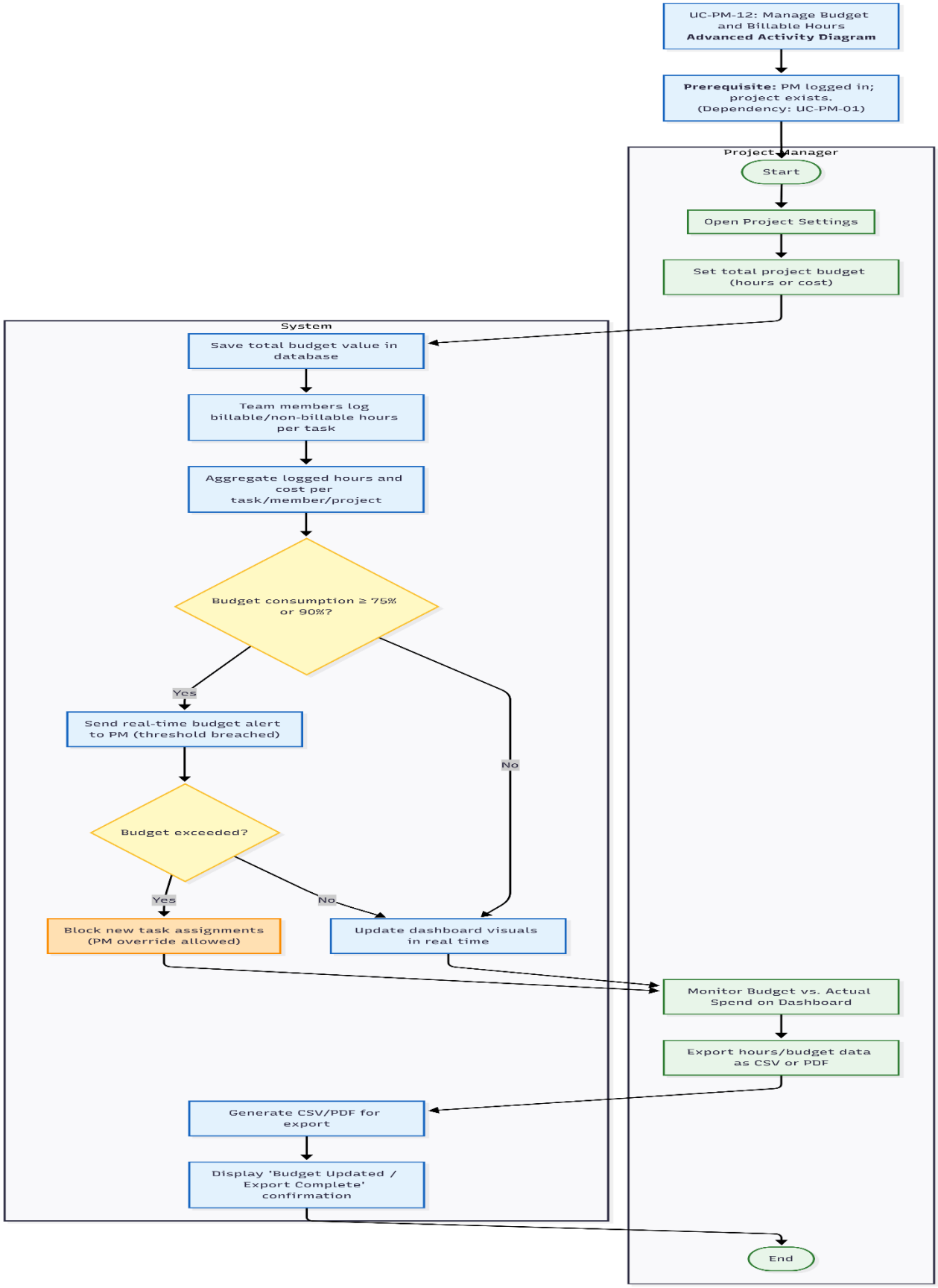

UC-PM-07: View Timeline / Gantt Chart
Advanced Activity Diagram
Prerequisite: PM logged in; tasks have valid start and due dates. (Dependencies: UC-PM-03, UC-PM-05)
Project Manager
View Tasks plotted across calendar axis
Drag Start or End of a Task Bar to adjust dates
Start
Navigate to 'Timeline / Gantt' View
System
Adjusted Date violates Dependency?
Yes
No
Save Updated Dates
Warn PM of Conflict Reject or Adjust Automatically
Load Task Data with Dates and Dependencies
Recalculate Critical Path and Dependencies
Any Task Missing Dates?
Yes
No
Confirm Successful Save
Prompt PM: 'Add Dates to Tasks to proceed'
Render Timeline / Gantt View with:
• Task Bars
• Dependencies (Arrows)
• Critical Path Highlight
End

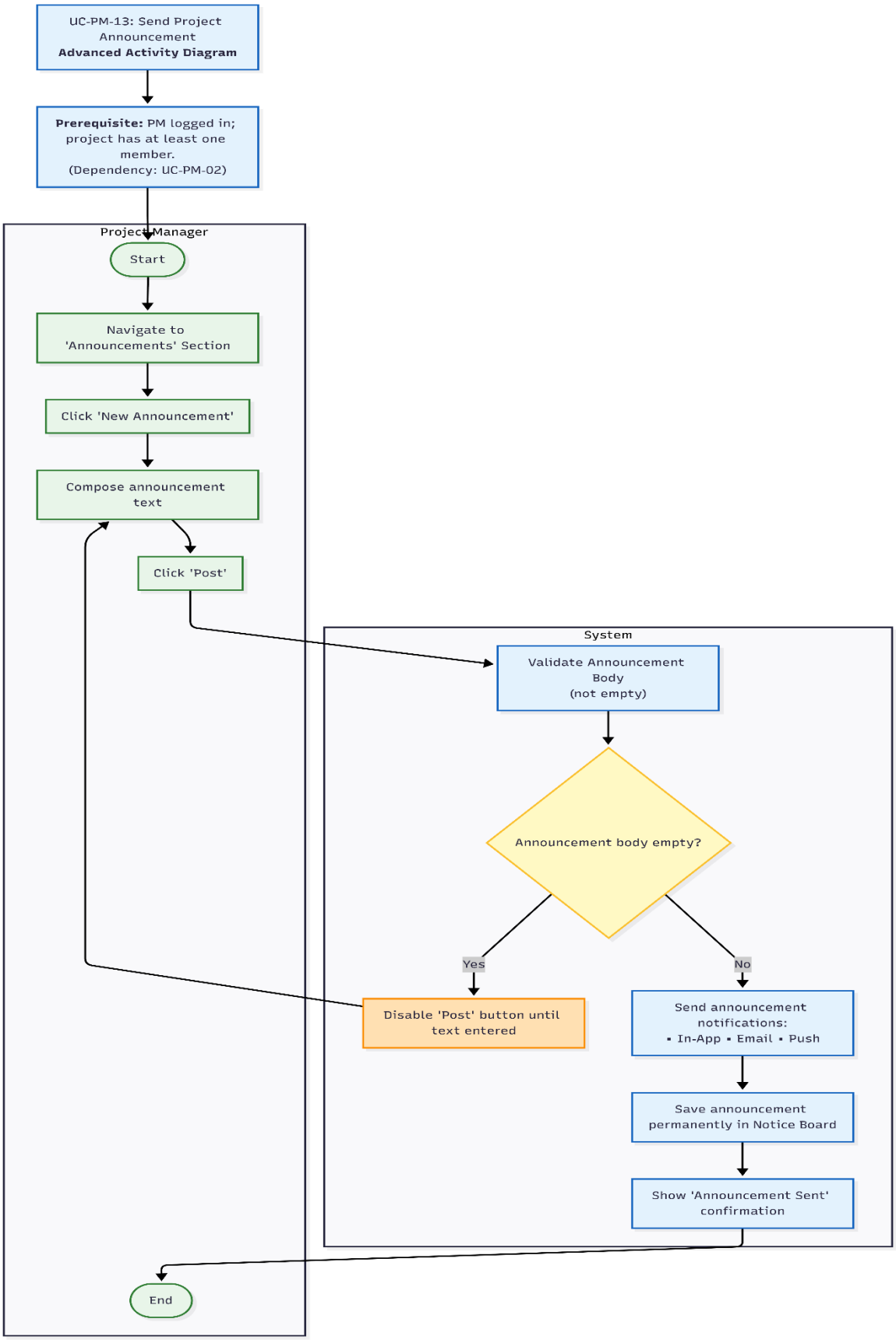

UC-PM-08: Monitor Team Workload
Advanced Activity Diagram
Prerequisite: PM logged in; project has members with assigned tasks. (Dependencies: UC-PM-02, UC-PM-03)
Project Manager
View workload bars per team member showing % capacity used
Start
Drag Task from one member to another to rebalance work
Navigate to 'Workload' Section of Project
System
Filter workload by sprint or date range
Load task assignments per member
Reassignment pushes target >100% capacity?
Yes
No
Calculate workload % against configured weekly capacity
Show warning before confirming reassignment
Update task ownership and recalc workloads in real time
Render color-coded workload bars:
• Green = Healthy
• Yellow = Near Limit
• Red = Overloaded
Broadcast updates to all project members
Show message: 'Workload Updated Successfully'
End


UC-PM-09: Generate Tasks from Meeting Audio
Advanced Activity Diagram
Prerequisite: PM logged in; project exists. (Dependencies: UC PM 01, UC PM 03)
Project Manager
Start
Navigate to 'Meetings' → select 'New Meeting Record'
Upload audio file (MP3/WAV/M4A) or record directly
System
Validate file format and size (≤ 2h audio)
File format supported?
No
Yes
Show 'Unsupported File Format' error
Transcribe audio with speech-to-text engine
Audio quality sufficient?
Good
Low
Flag transcription warning and allow manual edits
AI analyzes transcript to detect action items (Task, Owner, Priority, Deadline)
Show suggested tasks to PM for review
Review transcribed text and AI-suggested action items
Alternative: Approve all suggestions at once
Approve / Edit / Reject tasks
Create approved tasks in project backlog
Generate meeting summary (Decisions, Risks, Tasks, Next Steps)
Store audio, transcript, and summary in project data store
Notify PM: 'Tasks Created Successfully'
End


UC-PM-10: View Project Health Dashboard
Advanced Activity Diagram
Prerequisite: PM logged in; project has active tasks and members. (Dependencies: UC-PM-01, UC-PM-03)
Project Manager
Start
Navigate to Project Dashboard
System
Fetch real-time project metrics from database
Sufficient data available?
Yes
No
Compute Health Score from: Completion rate, Overdue tasks, Workload, Budget
Render Dashboard with live metrics and charts
Generate Suggested Corrective Actions
Auto-refresh dashboard data in real time
Prompt PM: 'Add tasks and members to generate dashboard'
View Health Indicators: Deadline Risk • Team Workload • Task Completion • Budget • Overall Score
Click indicator to view detailed breakdown
Review suggested corrective actions (e.g. Reassign tasks)
End


UC-PM-11: Generate Progress Report
Advanced Activity Diagram
Prerequisite: PM logged in; project has logged activity. (Dependency: UC-PM-10)
Project Manager
Start
Navigate to 'Reports' Section
Click 'Generate Report'
Select Report Type (Weekly / On-Demand) and Date Range
System
Validate that data exists in selected range
Data exists for selected period?
No
Yes
Show message: 'No activity in selected range'
Compile report data: Tasks Completed, Remaining, Hours, Budget, Risks
Generate PDF preview and online share link
Optional: Schedule automatic weekly report email
Valid email address for scheduling?
Preview generated report
No
Yes
Export Report as PDF or Copy Shareable Link
Show 'Invalid Email' warning
Save recurring weekly email schedule
Confirm 'Report Generated Successfully'
End


UC-PM-12: Manage Budget and Billable Hours
Advanced Activity Diagram
Prerequisite: PM logged in; project exists. (Dependency: UC-PM-01)
Project Manager
Start
Open Project Settings
Set total project budget (hours or cost)
System
Save total budget value in database
Team members log billable/non-billable hours per task
Aggregate logged hours and cost per task/member/project
Budget consumption ≥ 75% or 90%?
Yes
No
Send real-time budget alert to PM (threshold breached)
Budget exceeded?
Yes
No
Block new task assignments (PM override allowed)
Update dashboard visuals in real time
Monitor Budget vs. Actual Spend on Dashboard
Export hours/budget data as CSV or PDF
Generate CSV/PDF for export
Display 'Budget Updated / Export Complete' confirmation
End


UC-PM-13: Send Project Announcement
Advanced Activity Diagram
Prerequisite: PM logged in; project has at least one member.
(Dependency: UC-PM-02)
Project Manager
Start
Navigate to 'Announcements' Section
Click 'New Announcement'
Compose announcement text
Click 'Post'
System
Validate Announcement Body
(not empty)
Announcement body empty?
Yes
No
Disable 'Post' button until text entered
Send announcement notifications:
▪ In-App ▪ Email ▪ Push
Save announcement permanently in Notice Board
Show 'Announcement Sent' confirmation
End